

Near-Isothermal Compressed Air Energy Storage: Numerical and Experimental Investigation of a Plate Heat Exchanger as Compressor

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Abstract

This study investigates the potential of near-isothermal compressed air energy storage (nI-CAES) through a new approach that uses a plate heat exchanger as a compressor to minimize temperature differences during compression. An overview of major energy storage technologies positions CAES as a potential frontrunner for inter-seasonal and kilowatt-scale applications, necessitating the design of more efficient compression mechanisms. Analytical and numerical models were developed in order to estimate the feasibility of the planned design based on the principles of conservation of energy and mass. The analytical model included a semi-infinite axial thermal boundary layer model, a finite-domain gas-to-plate lateral heat transfer model, and a model of the wall response. The numerical model employed mass flow rate and heat flux functions to predict the performance of the system, and these were compared with experimental results. The model could simulate global thermodynamic trends in pressure, mass flow rate, temperature, and heat flux. However, assumptions of ideal gas behavior and simplified handling of gravitational potential caused pressure gradients to be underestimated and heat transfer efficiency to be overestimated. Experimental verification confirmed that a plate heat exchanger can be used as a compressor. Future work should explore the replacement of the compressor with a water pump for increased efficiency and versatility, and further develop the system to an overall compression–expansion nI-CAES system, as well as building advanced two-dimensional models in order to learn more about heat transfer mechanisms.

1 . Introduction

As the global energy sector transitions toward net zero emissions, the integration of intermittent renewable and periodic sources such as solar and wind power has highlighted the critical need for long-duration energy storage solutions since supply peak and demand peak are mismatched [1], [2]. Among these, inter-seasonal to inter-annual energy storage is essential to bridge seasonal mismatches between supply and demand by storing energy across months from low demand months to high demand months and provide necessary resilience to avoid blackouts during prolonged periods of low output from renewables (also known as Dunkelflaute episodes). For energy storage, there are some conventional storage methods and each one of them faces challenges in scalability, cost, location dependency, or duration.

Pumped hydro-electricity storage (PHES) is the most mature energy storage method with a total installed capacity of 158 GW in 2023, which represents more than 90% of global grid-scale energy storage capacity [3]. In a PHES plant, electricity is consumed to pump water from a low-altitude position to a higher position to store energy as potential energy. To recover the stored energy, the high-potential water flows downwards through a set of turbines, thus generating electricity [1]. Hydrogen electricity storage (HES) is a low-TRL method to store high-pressure hydrogen or other hydrogen-involved compounds to achieve high energy density, and the source of hydrogen is the main concern [4], [5], [6]. Electro-chemical electricity storage (ECES) is a well-developed energy storage solution for both grid-scale storage and commercial storage, but the significant self-discharge rate makes ECES

candidates unsuited for inter-seasonal to inter-annual energy storage [7], [8].

Thermo-mechanical electricity storage (TMES) solutions consist of storing electricity as both mechanical and thermal energy [8]. TMES options include several technologies such as pumped thermal electricity storage (PTES), liquid air electricity storage (LAES), and compressed air electricity storage (CAES) [9], [10], [11].

However, due to the inter-seasonal energy duration and energy capacity requirements, only part designs of PHES, HES, and CAES can store energy for about 2 months and hold in GWh scale. To specify more, PHES is the most mature storage with long discharge duration but low energy density (2 kWh/m³) and high site requirements [3]. HES offers high density (600 kWh/m³) and long duration yet suffers from high costs (~5000 USD/kW) [5], [6]. CAES presents a balanced solution with moderate density (8 kWh/m³), lower geological constraints, and lower cost (~1000 USD/kW), making it a strong candidate for inter-seasonal storage [11], [12].

CAES has evolved into two main configurations: diabatic CAES (D-CAES) and adiabatic CAES (A-CAES). In a CAES plant, electricity is used to drive a compressor and store high-pressure air in a tight reservoir, typically a salt cavern. During the near-adiabatic compression process, the gas temperature rises due to the density change what will make compression harder, and high amounts of heat is required to make air expansion more efficient.

D-CAES systems use extra cooler and fossil fuels to take and give heat to compression and expansion, respectively. The first commercial D-CAES plant, the Huntorf facility in Germany, stores approximately 310,000 m³ of air at pressure ranging from 44 to 70 bar achieving a round-trip

Nomenclature

Symbols

$\dot{m}$	mass flowrate in the system [m^3/s]	o	input conditions at the water tank
ρ	density of water at the standard condition [kg/m^3]	p	pipeline
$c_{P,i}$	heat capacitive for constant pressure [$\frac{\text{J}}{\text{kg K}}$]	r	needle valve
A	cross-section area [m^2]	g	gas in the test section
A_i	cross-section area of the inlet plate heat exchanger [m^2]	b	the bulk
P_o	ambient pressure [bar]	s	wall
P	air pressure in the test section unit [bar]	x	the vertical axis [m]
h_o	water level at the pressurized reservoir [m/s]	y	the horizontal axis, dependent on time
h	water level at the test section unit [m/s]		
u	fluid velocity at the test section unit [m/s]		
u_o	fluid velocity at the pressurized reservoir [m/s]		
u_p	fluid velocity inside the pipeline [m/s]		
k_i	loss coefficient		
K	overall system loss coefficient		
R	Reynold number		
Z_i	ratio of the change in wall temperature		
T_i	temperature [K]		
t	time [s]		
a_o	thermal diffusivity [m^2/s]		
Q	heat flux [$\frac{\text{J}}{\text{s m}^2}$]		
g	Gravity [m/s^2]		

Greek symbols

η	vertical position change variable
τ	time change variable
ξ	vertical position change variable
σ	time change variable
θ_i	ratio
φ_i	heat flux [$\frac{\text{J}}{\text{s m}^2}$]

Subscripts

$i = 1, 2, 3, 4$	For $k, i = 1, 2, 3, 4$ stands for pipeline elbow loss coefficient at the reservoir side and at the test section side, and test section inlet loss respectively
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Abbreviations

CAES	compressed air energy storage
nI-CAES	near-isothermal compressed air energy storage
A-CAES	adiabatic compressed air energy storage
D-CAES	diabatic compressed air energy storage
PTES	pumped thermal electricity storage
LAES	liquid air electricity storage
PHES	Pumped hydro-electricity storage
HES	Hydrogen electricity storage
ECES	Electro-chemical electricity storage
TMES	thermo-mechanical electricity storage
FI	flowrate indicator
PI	pressure indicator
TI	temperature indicator

efficiency of around 42% [11], [13]. However, due to the nature of this system, high amount of heat is not only wasted, but it also requires extra heat from fossil fuels, so D-CAES creates high carbon emissions with low round-trip efficiency.

A-CAES has a thermal storage system to store the generated heat during compression process and reuse them during the expansion process, but it aims the reuse of heat generated which is converting the mechanical work loss to be the making up of it. This mechanical work loss only takes a part of whole energy used, which leads to round-trip efficiencies of up to 65-70% [11], [14], [15]. More importantly, A-CAES plants suffer from significant self-discharge rates due to hard-to-avoid heat losses from the thermal stores, which makes it a sub-optimal solution for (very) long-duration storage.

To remedy this problem, near-isothermal compressed air electricity storage (nI-CAES) aims to keep the temperature constant during compression and expansion processes and solve the thermal management issues limiting the potential of A-CAES.

In order to minimize the temperature changes during compression and expansion processes, the overall air-to-ambient thermal resistance must be minimal. In other terms, the product of the heat transfer area and overall heat transfer coefficient must be maximized. This means that a trade-off must be found between round-trip efficiency (i.e., a measure of how close to isothermal compression and expansion processes are) and power density (i.e., how large the compression/expansion machinery is for a given power rating) [12].

During gas compression, there are many ways to minimize temperature changes including heat transfer to an external medium, multi-stage compression with intercooling, and slow compression. In this report, the design of including heat transfer to an external medium will be considered.

Liquid water is a suitable medium for several reasons [16]. First, its relatively high thermal conductivity allows for efficient heat transfer. Second, its high specific heat capacity enables it to absorb and store more thermal energy. Finally, due to its high density compared to air, it can store more thermal energy per unit volume. In addition, water is incompressible, which simplifies its collection, storage, and reuse within the system. There are various ways to transfer heat from the air to a water body, and they all aim to increase contacting time and area between air and water.

One promising method is water spraying, which involves injecting water droplets into the air during compression with a (typically solid reciprocating) piston. This maintains continuous and effective heat transfer. The compressed air and water mixture is then separated, allowing storage of high-pressure air and water independently. Research from Hua Chen

shows that by integrating the water spraying injection can improve the round-trip efficiency of normal piston compression can be improved from 50% to 76% [17]. While this design significantly enhances heat transfer from the air, further improvements are possible by optimizing heat transfer involving the pump and piston components.

The liquid piston nI-CAES design proposed in this work not only considers the heat transfer from air but also from the piston and compressor chamber. In the configuration, a liquid such as water functions as a piston, displacing air upward through the compressor chamber. This setup facilitates direct heat exchange both at the air-liquid interface and at the liquid-wall interface of the chamber [18].

However, the effective air-liquid interface area in this design is smaller than that achieved by the water spray method, which limits the potential improvement in round-trip efficiency.

To enhance the air-liquid interface and overall thermal management, a heat exchanger can be integrated into the compressor chamber. This approach simultaneously improves heat dissipation from both the air and the surrounding compressor structure during compression.

Despite the theoretical advantages of liquid piston nI-CAES systems with integrated thermal management, both simulation-based analyses and experimental investigations are largely lacking in the existing literature. This lack of exploration limits the current understanding of heat recovery and reuse in integrated CAES designs, thereby hindering the advancement of more efficient, compact, and sustainable energy storage technologies.

The aim of this study is to advance the design of bidirectional liquid-piston compression/expansion machines based on repurposed plate heat exchangers to develop affordable nI CAES solutions. In particular, we aim

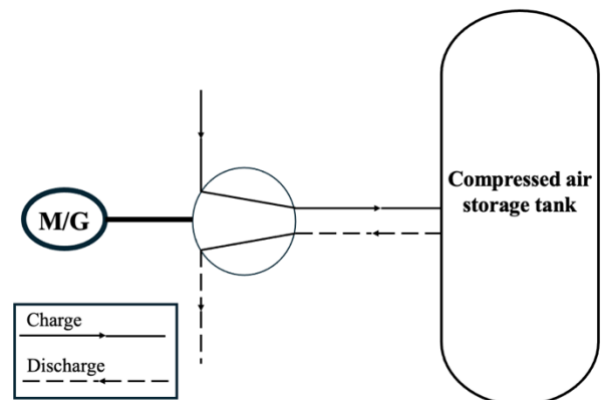


Figure 1: Isothermal compressed air electricity storage charging and discharging process by using a reversible compression/expansion device.

at deriving simplified yet accurate analytical models of such near-isothermal reversible engines and verify their accuracy by comparing model predictions with high-fidelity experimental measurements on an early-stage prototype.

The structure of the paper is as follows. Section 2 presents in more details the working principle and inner workings of the proposed nI-CAES system. The analytical, experimental and numerical methods are described in Section 3. Section 4 reports the simulation results and compares them with generated experimental data. Finally, Section 5 summarizes the key conclusions and implications of the study.

A reversible compressor functions as a motor during charging process, using electricity to pressurize air and store pressurized air in a storage tank. During this process, heat is generated. This reversible compressor operates as a generator during the discharging process by expanding the stored air. In I-CAES, shown in **Figure 1**, the reversible compressor can simultaneously operate as both a motor and a generator, enabling continuous expansion and compression.

Liquid piston technology as mentioned earlier, provides a potential to operate as a reversible compressor. In **Figure 2**, during (a) the compression process, water is pumped to displace the air, thereby reducing its volume, increasing its pressure, and releasing heat. During (b) the expansion process, the pressurized air is released to drive the liquid piston through a turbine to generate electricity, while additional heat is supplied by the heat recovered during compression via the liquid piston. This closed-system design offers advantages by reducing air leakage and providing improved heat transfer compared with other piston-based reversible compressor designs.

2 . System description

As mentioned in the introduction, the fundamental liquid piston technology demonstrates poor air–liquid interfacial and heat transfer performance, despite its advantages. A plate heat exchanger integrated with a liquid piston is repurposed as a compressor to overcome these limitations.

In this design, the plate heat exchanger possesses additional heat transfer nodes, including the air–liquid interface, air–wall interface, liquid–wall interface, wall–ambient air interface, and an additional wall–adjacent plate water interface, compared with the fundamental liquid piston design.

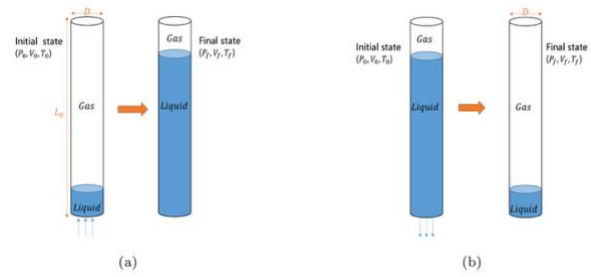


Figure 2: Operations of liquid piston during (a) compression and (b) expansion process. Adapted from El Mehdi Gouda, Review on Liquid Piston technology for compressed air energy storage.

This additional wall–adjacent plate water interface potentially enhances heat transfer.

In the **Figure 3**, it illustrates how liquid piston is applied in heat exchangers. During the compression process, the water contained in Heat exchanger 1 (at near ambient pressure) is circulated by the pump to flow into, and compress the air in, Heat exchanger 2. Once the supply pressure is reached, the high-pressure is delivered and the process repeated with Heat Exchanger 2 draining and Heat Exchanger 1 filling. During the expansion process, water in Heat Exchanger 2 is pressurized by the incoming high-pressure air, which drives the water to Heat Exchanger 1 through a hydraulic turbine, thus generating useful work and electricity (via a generator). For temperature management, the traditional liquid piston technology has heat transfer nodes at gas–liquid interface, gas–wall interface, water–wall interface, and wall–ambient air interface. For this design, the wall–ambient air interface heat transfer node is replaced by wall–water interface heat transfer node.

To evaluate this design capability in managing temperature variations, an analytical approach, based on James C. Keck’s two-dimensional formulations of energy and mass conservation, is applied to model heat transfer at the axial thermal boundary layer, the gas-to-plate interface, and the wall’s thermal response [19]. The experimental setup and objectives are then introduced. Using the computational tool Python, system responses under different conditions are predicted through numerical methods, and the predicted results are subsequently compared with the experimental measurements in the later discussion section.

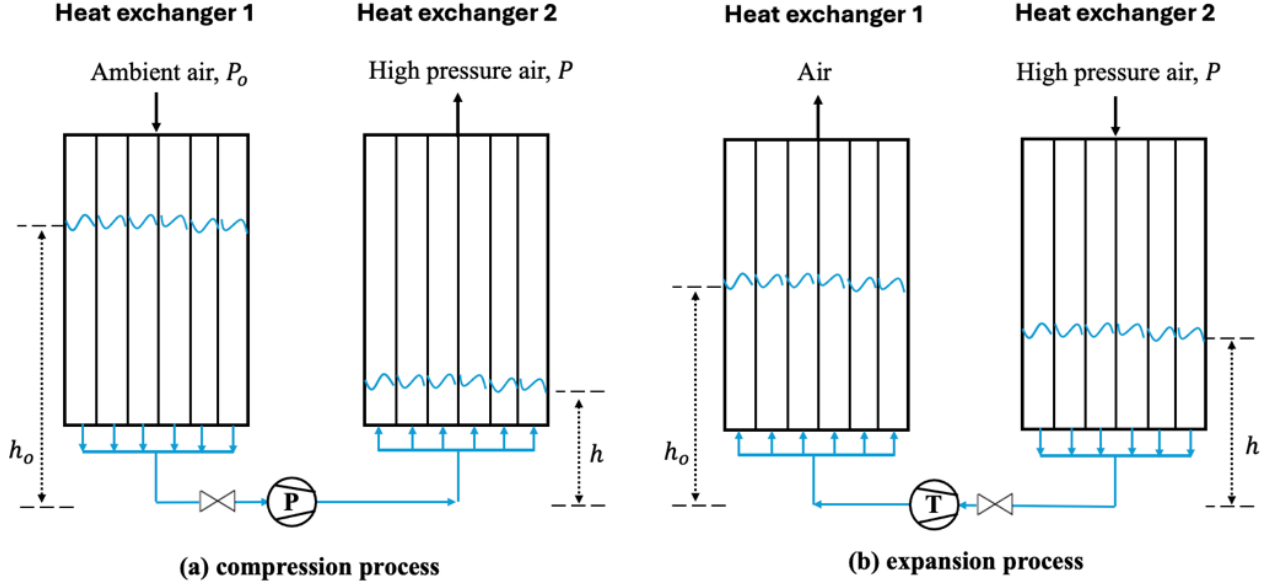


Figure 3: Liquid piston with plate heat exchanger (a) compression and expansion process include a hydraulic head and reversible pump-turbine (P / T).

3 . Methods

3.1. Analytical methods

The conjugate pressure-driven gas-to-wall heat transfer occurring in liquid-piston devices is a transient problem that can be described by a set of three governing equations, namely: Eq. (1), a diffusion-advection heat equation to predict the temperature field in the gas phase as a function of the volume-pressure trace; Eq. (2), a continuity equation to ensure conservation of mass; and Eq. (3) a diffusion heat equation in the solid phase:

$$\rho_g c_{P,g} \left(\frac{dT_g}{dt} + \vec{v}_g \cdot \vec{\nabla} T_g \right) = \frac{dP_g}{dt} + \vec{\nabla} \cdot (k_g \vec{\nabla} T_g); \quad (1)$$

$$\frac{\partial \rho_g}{\partial t} + \vec{\nabla} \cdot (\rho_g \vec{v}_g) = 0; \text{ and} \quad (2)$$

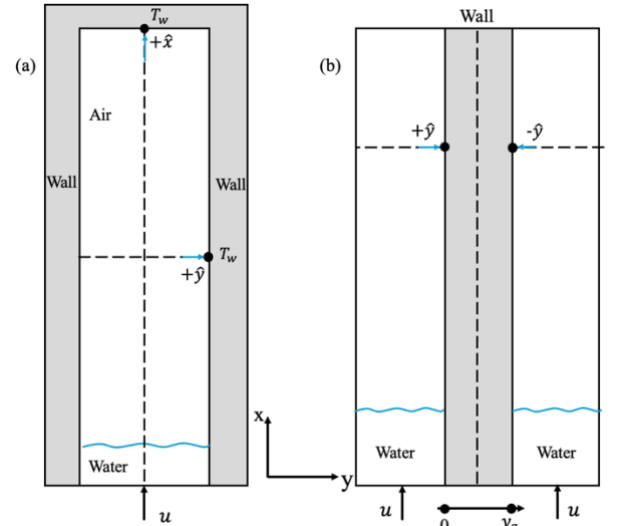
$$\rho_s c_{P,s} \frac{dT_s}{dt} = \vec{\nabla} \cdot (k_s \vec{\nabla} T_s). \quad (3)$$

Below, we derive 1D analytical solutions to these equations based on the following assumptions:

1. System's gas follows the ideal gas law.
2. Liquid is incompressible.
3. No corner effect.
4. The gas external compression is isentropic.
5. In the numerical model, the inlet air pressure is assumed to be constant.

3.1.1. Semi-infinite axial thermal boundary layer

As a starting point, let us inspect the semi-infinite solution to the 1D axial (i.e., vertical) thermal boundary layer derived by Keck, which will guide us when deriving the analytical solution for the



lateral gas-to-plate heat transfer. In the vertical direction, as **Figure 4 (a)** shown, axial heat transfer happens at both gas to plate wall head and gas to

Figure 4: Scheme of 2 dimensions with defined direction, boundary conditions, and symmetry axis. Scheme (a) is defining the heat transfer from air to water, and air to wall. Scheme (b) is defining the heat transfer from water to wall in two plates.

liquid piston. In this discussion, the 1D model on vertical axis, the x-axis, can be present as:

$$\rho_g c_{P,g} \left(\frac{dT_g}{dt} + u_g \frac{dT_g}{dx} \right) = \frac{dP_g}{dt} + \frac{d}{dx} \left(k_g \frac{dT_g}{dx} \right), \quad (4)$$

where k_g the gas thermal conductivity is defined as:

$$k_g = k_{g,0} \frac{T_g}{T_{g,0}}. \quad (5)$$

The conservation of mass equation can be modified to eliminate the advection term:

$$\frac{d\eta}{dt} = \frac{\rho_g}{\rho_{g,0}} u_g = 0 \quad (6)$$

which leads to

$$\eta(x, t) = \int_0^x \frac{\rho_g}{\rho_{g,0}} \frac{1}{dx} \rightarrow \frac{1}{dx} = \frac{\rho_g}{\rho_{g,0}} \frac{1}{d\eta}. \quad (7)$$

And the mass conservation in x-axis is

$$\rho_{g,0} c_{P,g} \frac{dT_g}{dt} = \frac{\rho_{g,0}}{\rho_g} \frac{dP_g}{dt} + \frac{d}{d\eta} \left(k_g \frac{\rho_g}{\rho_{g,0}} \frac{dT_g}{d\eta} \right). \quad (8)$$

By using ideal gas law

$$P_g = \rho_g R T_g, \quad (9)$$

based on the assumption of the gas external isentropic compression to the boundary layer:

$$\frac{dP_g}{dt} = \frac{P_g c_{P,g}}{R T_b} \frac{dT_b}{dt}. \quad (9)$$

Then,

$$\rho_{g,0} c_{P,g} \frac{dT_g}{dt} = \frac{P_g c_{P,g}}{R T_b} \frac{dT_b}{dt} + \frac{d}{d\eta} \left(k_g \frac{\rho_g}{\rho_{g,0}} \frac{dT_g}{d\eta} \right). \quad (10)$$

With arrangements, the equation is

$$\rho_{g,0} c_{P,g} T_b \frac{dT_g}{dt} = \frac{d}{d\eta} \left(k_g \frac{\rho_g}{\rho_{g,0}} \frac{dT_g}{d\eta} \right). \quad (11)$$

By defining a time change variable, which is

$$\tau = \int_0^t \frac{P_g}{P_{g,0}} \frac{1}{dt}. \quad (12)$$

By reapplying the ideal gas law back, the equation can be further changed like

$$\rho_{g,0} c_{P,g} T_b \frac{dT_g}{dt} = T_b k_g \frac{\rho_g}{\rho_{g,0}} \frac{d^2 T_g}{d\eta^2 T_b}. \quad (13)$$

With arrangements

$$\frac{d\theta_x}{d\tau} = a_0 \frac{d^2 \theta_x}{d\eta^2}, \quad (14)$$

where

$$a_0 = \frac{k_g \rho_g P_{g,0}}{\rho_{g,0}^2 c_{P,g} P_g}, \text{ and} \quad (15)$$

$$\theta_x = \frac{T_g}{T_b}. \quad (16)$$

T_b is bulk temperature can be calculated as:

$$T_b = T_{g,0} \left(\frac{P_g}{P_{g,0}} \right)^{\frac{\gamma-1}{\gamma}}, \text{ where} \quad (17)$$

$$\gamma = \frac{c_{P,g}}{c_{v,g}}. \quad (18)$$

Now, the boundary conditions and initial conditions are settled to have:

$$\theta_x(\eta = 0, \tau) = \theta_w = \frac{T_w}{T_b}; \quad (19)$$

$$\lim_{x \rightarrow \infty} \theta_x = 1; \text{ and} \quad (20)$$

$$\theta_x(\eta, \tau = 0) = \theta_0 = \frac{T_{g,0}}{T_b}. \quad (21)$$

θ_0 is the initial temperature ratio, the vertical axis is semi-infinite.

By applying those conditions, the Laplace domain is

$$\hat{\theta}_x(\eta, s) = \frac{\theta_{g,0}}{s} + \frac{s \hat{\theta}_w(s) - \theta_0}{s} e^{-\eta \sqrt{\frac{s}{a_0}}}. \quad (22)$$

By doing inverse Laplace transform, the equation is

$$\theta_x(\eta, \tau) = 1 + \operatorname{erfc} \left(\frac{\eta}{2\sqrt{\tau a_0}} \right) * \frac{d\theta_w}{d\tau}, \quad (23)$$

where $*$ is the convolution product.

And the heat flux at the wall head and gas-liquid interface can be found by using Fourier law:

$$\varphi_x = \frac{\varepsilon_0}{\sqrt{\pi}} \frac{T_b^2}{T_{g,0}} \left(\frac{1}{\sqrt{\tau}} * \frac{d\theta_x}{d\tau} \right). \quad (24)$$

3.1.2. Finite-domain solution for the gas-to-plate lateral heat transfer

In horizontal direction, as **Figure 4 (a)** shown, symmetry y-axis, is dependent on time, the lateral heat transfer happens at gas to wall. In this discussion, the 1D model on y-axis, can be present as:

$$\rho_g c_{P,g} \left(\frac{dT_g}{dt} + \vec{v}_g \frac{dT_g}{dy} \right) = \frac{dP_g}{dt} + \frac{d}{dy} \left(k_g \frac{dT_g}{dy} \right). \quad (25)$$

The position change and time change variables are:

$$\xi = \int_0^y \frac{\rho_g}{\langle \rho_g \rangle} dy; \text{ and} \quad (26)$$

$$\sigma = \int_0^t \frac{P_{g,0}}{P_g} \left(\frac{\langle T_g \rangle}{T_{g,0}} \right)^2 dt. \quad (27)$$

$\langle \rho_g \rangle$ and $\langle T_g \rangle$ are mass average density and mass average temperature of air respectively. Then, the equation is

$$\frac{\partial \theta_y}{\partial \sigma} = a_0 \frac{\partial^2 \theta_y}{\partial \xi^2}, \text{ where} \quad (28)$$

$$\theta_y = \frac{T_g}{T_b}; \text{ and} \quad (29)$$

$$a_0 = \frac{k_g \rho_g P_{g,0} T_{g,0}^2}{\langle \rho_g \rangle P_{g,0} \langle T_g \rangle^2 c_{p,g}}. \quad (30)$$

The boundary conditions and initial conditions are settled to have:

$$\frac{\partial \theta_y}{\partial \xi} (\xi = \xi_R, \sigma) = 0; \quad (31)$$

$$\theta_y (\xi = 0, \sigma) = \theta_w; \text{ and} \quad (32)$$

$$\theta_y (\xi, \sigma = 0) = \theta_0. \quad (33)$$

By applying those conditions, the Laplace domain is

$$\hat{\theta}_y = \frac{1}{s} \left((s \hat{\theta}_w - 1) \frac{\cosh(\xi_R - \xi)p}{\cosh \xi_R p} + 1 \right), \quad (34)$$

where $p = \sqrt{\frac{s}{b_0}}$, by doing inverse Laplace transform to have the temporal domain:

$$\theta_y = \theta_0 + \frac{\partial \theta_w}{\partial \sigma} * (1 - N); \text{ and} \quad (35)$$

$$N = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(n+\frac{1}{2})(-1)^n \cos\left[(n+\frac{1}{2})\pi \frac{\xi_R - \xi}{\xi_R}\right]}{(n+\frac{1}{2})^2} e^{-(n+\frac{1}{2})^2 \pi^2 \frac{\sigma}{\xi_R^2}}. \quad (36)$$

where $*$ is the convolution product.

3.1.3. Wall thermal response

Wall thermal response is considered on horizontal dimension. For the plate heat exchanger with liquid piston, a plate wall heat transfer with both two-side lateral, which also be present in **Figure 4 (b)**, the energy conservation can be present as:

$$\frac{\partial \theta}{\partial t} = a_0 \frac{\partial^2 \theta}{\partial y^2}, \text{ where} \quad (37)$$

$$\theta = T_w - T_{w,0}, \text{ and} \quad (38)$$

a_0 is the wall thermal diffusivity.

The equation:

$$T_w = T_{w,0} + Z_w * \varphi_w \quad (39)$$

Is making the Z_w to be the ratio of the change in wall temperature and heat flux, and $*$ is the convolution product.

The Laplace domain is

$$\widehat{Z}_w = \frac{\hat{\varphi}}{k_g p} \left(\frac{e^{-y_a p}}{\sinh y_a p} \cosh y p + e^{-y p} \right), \quad (40)$$

where

$$p = \sqrt{\frac{s}{a_0}}. \quad (41)$$

By combining the θ_y function, the ratio of the change in wall temperature is

$$Z_w = \mathcal{L}^{-1} \left\{ \frac{\hat{\theta}_w(y=0)}{\hat{\varphi}} \right\} = \frac{\sqrt{y_a}}{k_g \sqrt{\pi t}} \left(1 + 2 \sum_{n=1}^{\infty} e^{-\frac{y_a^2 n^2}{a_0 t}} \right). \quad (42)$$

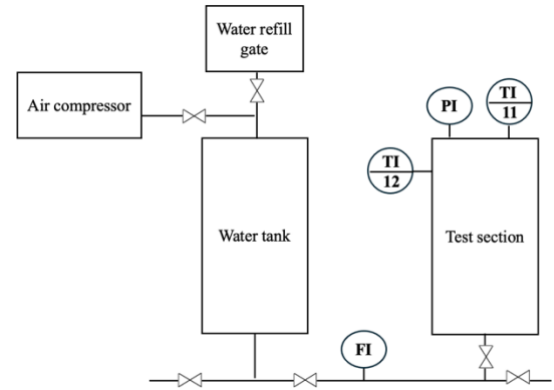


Figure 5: Scheme of experiment set-up. FI is the flowrate indicator to measure liquid flowrate, PI is the pressure indicator to measure air pressure in the test section compressor, TI 11 is the temperature indicator to measure air temperature, and TI 12 is the temperature indicator to measure wall temperature.

3.2. Experimental methods

The experiment is built by Yitong Han in the same lab under the supervision of Dr. Sapin Paul. The idea of experiment is building up one plate heat exchanger compressor with liquid piston which only considering compression discussion. In the setup, shown in **Figure 5**, an air compressor is set instead of pump to provide kinetic energy of liquid piston. Water in water tank is pushed to the test section unit. The test section unit is a plate heat exchanger, and the pushed water works as liquid piston to compress air inside of the exchanger. Series of valve is set to meet switching, safety, and maintenance need.

Experiment is intended to measure mass flowrate, air temperature, wall temperature,

temperature change, pressure, and heat flux under different conditions.

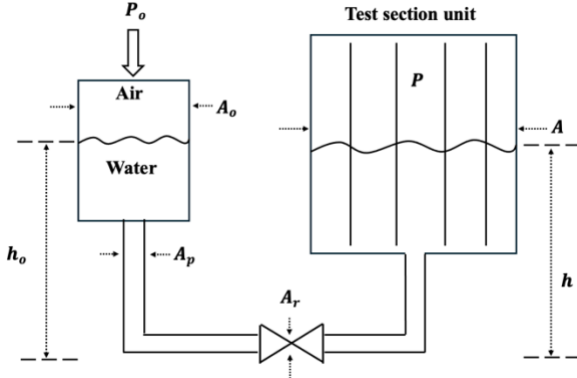


Figure 6: Schematic of the experimental setup put together to investigate a single-stroke compression process in a plate heat exchanger repurposed as a liquid-piston compressor (denoted as test section in the figure).

3.3. Numerical model

3.3.1. Pressure loss in the system

An illustration of this step modelling is in **Figure 6**. To observe how a streamline behaves that starts from the pressurized reservoir to the compressor, the Bernoulli equation can be applied in form of

$$P_o + \rho g h_o + \frac{\rho u_o^2}{2} = P + \rho g h + \frac{\rho u^2}{2} + k \frac{\rho u_p^2}{2} \quad (43)$$

where $K \frac{\rho u_p^2}{2}$ is the overall system head losses with

$$K = k_r + k_p + \sum_{i=1,2,3,4} k_i. \quad (44)$$

With arrangements, the equation is

$$\frac{\rho u^2}{2} = (P_o - P) + \rho g(h_o - h) + \frac{\rho u_o^2}{2} - K \frac{\rho u_p^2}{2}. \quad (45)$$

For those velocities, they can be converted to each other by using continuity equations:

$$A_p u_p = A_o u_o; \text{ and} \quad (46)$$

$$A u = A_o u_o. \quad (47)$$

At the point of compressor, the velocity of water u is defined as

$$u = \sqrt{2 \frac{\Delta P + \rho g \Delta h}{\rho \left(1 - \frac{A^2}{A_o^2} + k \frac{A^2}{A_p^2}\right)}}, \text{ where} \quad (48)$$

$$\Delta P = (P_o - P), \text{ and} \quad (49)$$

$$\Delta h = (h_o - h). \quad (50)$$

The water mass flowrate can be then present as:

$$\dot{m} = u A \rho = \sqrt{2 \frac{\rho (\Delta P + \rho g \Delta h)}{\frac{1}{A^2} - \frac{1}{A_o^2} + K \frac{1}{A_p^2}}}. \quad (51)$$

Set

$$C_1 = \frac{2 \rho (P_o - P - \rho g h_{\text{tank}} \frac{A_o}{A})}{\left(\frac{1}{A^2} - \frac{1}{A_o^2} + \frac{K}{A_p^2}\right)}; \text{ and} \quad (52)$$

$$C_2 = \frac{2 \rho^2 g (1 + \frac{A_o}{A})}{\left(\frac{1}{A^2} - \frac{1}{A_o^2} + \frac{K}{A_p^2}\right)}, \quad (53)$$

So

$$\dot{m} = \sqrt{C_1 + C_2 \dot{h}_o}. \quad (54)$$

The change of height of water level within time in the water tank is functioned as:

$$\dot{h}_o = \frac{\dot{V}}{A_o} = \frac{\sqrt{C_1 + C_2 \dot{h}_o}}{\rho A_o}. \quad (55)$$

Which is unsolvable by considering it as non-linear function, so the difference in height term is eliminated to have:

$$\dot{m} = \sqrt{2 \frac{\rho (P_o - P)}{\frac{1}{A^2} - \frac{1}{A_o^2} + k \frac{1}{A_p^2}}}. \quad (56)$$

the mass flowrate of water now is simply varied with pressure difference.

3.3.2. Heat flux

According to the first law of thermodynamics, the global heat flux in the system during compression process can be expressed as:

$$\dot{Q} = \dot{m} c_p \frac{dT}{dt} - \dot{V} \frac{dP}{dt} \quad (57)$$

By having the pressure loss and global heat flux equation, the numerical model can be built now in Python to predict experiment results.

4 . Results

This section presents the results obtained from both numerical modeling and experimental measurements. The numerical results provide predictive insights, while the experimental results serve to validate and assess the reliability of the model.

4.1. Numerical method predicted results

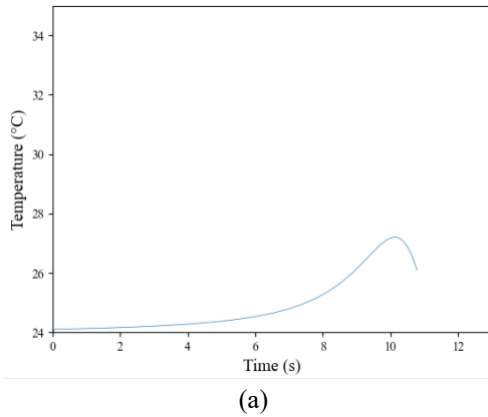
By looking the air temperature at the test section, in **Figure 7**, with setting constant input pressure at 10 bar, the original temperature is benchmarked with the experiment environmental temperature. From time 0 to 9 second, the slope of temperature rises and meet the peak at 303 K at 10, then slope is turning to negative. It happens similarly to pressure with time. Slope of pressure gradually increasing until 10 second.

By looking at equation 56,

$$\dot{m} = \sqrt{2 \frac{\rho(P_o - P)}{\frac{1}{A^2} - \frac{1}{A_o^2} + k \frac{1}{A_p^2}}} \quad (56)$$

By fixing the inlet pressure, the compressed air's pressure is gradually increasing and making liquid flowrate to be decreased. After compressed air's pressure meeting the inlet pressure, flowrate is zero and compression process stop. Temperature of compressed air is then decreasing by heat transferring with wall and water.

If instead of making inlet pressure to be constant, mass flowrate is set as constant in **Figure 8**. With



constant mass flowrate, the compression process can have theoretically higher limit on compressed air's pressure. In this case, temperature is not decreasing but increasing exponentially.

However, making mass flowrate constant is impossible. To solve it, by setting higher input pressure at 15 bar, compression process can be extended longer by obtaining more constant mass flowrate by comparing with constant input pressure at 10 bar.

Figure 9 illustrates the new condition's result. Now, compressed air temperature and pressure is now exponentially increasing with time.

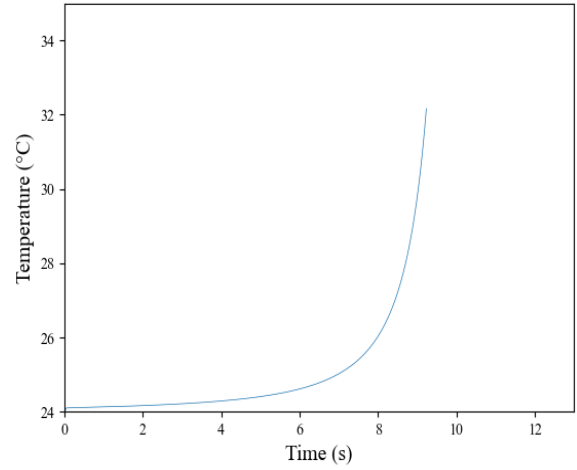


Figure 8: Test section air temperature under constant flowrate.

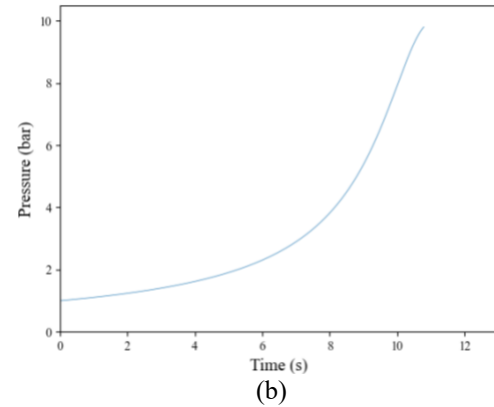


Figure 7: Test section air: (a) temperature; and (b) pressure with time under constant input pressure at 10 bar.

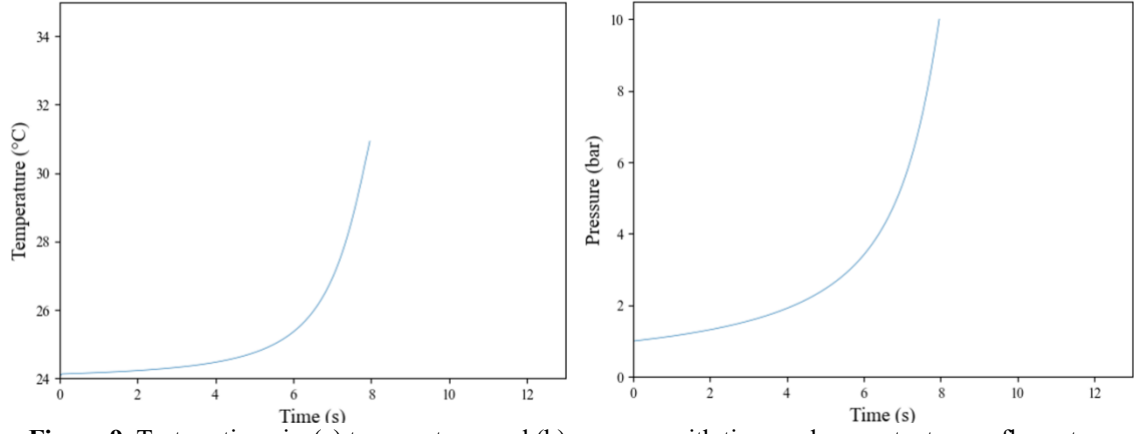


Figure 9: Test section air: (a) temperature; and (b) pressure with time under constant mass flowrate.

In **Figure 10**, two heat transfer coefficients include adiabatic scenario and isothermal scenario are selected to obtain wall temperature behavior for constant input pressure. Under adiabatic scenario, heat is transfer from wall then to ambient medium which indicates a lower heat transfer coefficient and higher wall temperature change by comparing with isothermal scenario's result. In isothermal scenario, heat is transferred efficiently which making wall temperature change with a bit. It is a promising result under concept of nI-CAES to minimize temperature change of air.

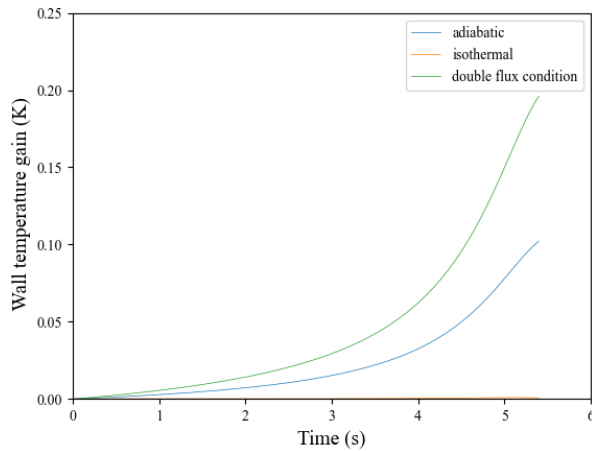


Figure 10: Wall temperature change during time under adiabatic, isothermal, and double flux scenarios.

4.2. Experimental results and model verification

4.2.1. Pressure

For Experiments 1, 3, and 4 in **Figure 11**, the initial pressure matches that of the numerical model at 1 bar. However, Experiment 2 shows a higher initial pressure of approximately 1.3 bar, likely due to measurement or initialization errors. Since the deviations in the other three experiments are relatively small, the measurement error can be considered negligible. If the difference originates

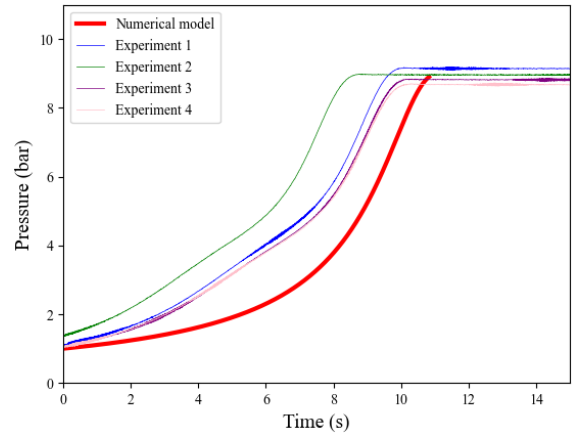


Figure 11: Pressure change during time. Experiment 1, 2, 3, and 4 are repeated experiments under same experiment conditions.

from initialization error, then the 0.3 bar discrepancy should be critically compared with the other results. Although the initial pressure of Experiment 2 is higher, its final (equilibrium) pressure is not greater than that of Experiment 1, but it does exceed the final pressures of Experiments 3 and 4.

One important aspect requiring discussion is that the inlet pressure in the experimental setup was not constant across different trials. The inlet pressure varied, with a maximum deviation of about 0.4 bar from the nominal 10 bar. If the test section pressure was initialized incorrectly under a lower inlet pressure, the entire compression process became inefficient, leading to a reduced terminal pressure and lower temperature, as shown in **Figure 13**.

Beyond the initial pressure, the pressure slope in the different experiments can also be analyzed. Excluding Experiment 2, which is regarded as affected by systematic error, Experiments 1, 3, and 4 display similar slope behavior. In Phase 1 (0–8 s), the pressure gradually increases. During this period, the compression experiences relatively low resistance, with a small pressure ratio between inlet and compressed air. Heat transfer is effective in moderating air expansion. In Phase 2 (8–9 s), the resistance rises sharply due to the limited inlet

pressure, producing a marginal effect. This corresponds with the faster rise in test section air temperature observed in **Figure 13**.

In Phase 3 (after 9 s), the test section pressure approaches the system limit. It cannot reach the inlet pressure because of energy conservation, kinetic losses, and thermodynamic losses. From a mass and energy balance perspective, the compressor power is first converted into the kinetic energy of water in the pipes, and then into the potential energy of the compressed air in the test section. Energy losses occur during this transfer, including both heat dissipation and kinetic energy loss, leading to increased air temperature and decreased water mass flow rate. As a result, the achievable pressure is limited to around 8.5 bar in these experiments.

The numerical model produces a qualitatively similar but more idealized pressure evolution over time. Starting from the same initial pressure, the numerical results show a slower increase: the pressure rises by only 2 bar within 8 seconds, compared with the 5-bar rise observed in the experiments. The model neglects gravitational potential effects because of the nonlinearity of the water level equations, effectively making the simulated compression appear more efficient. As shown in Equation (51):

$$\dot{m} = uA\rho = \sqrt{2 \frac{\rho(\Delta P + g\rho\Delta h)}{\frac{1}{A^2} - \frac{1}{A_0^2} + K \frac{1}{A_p^2}}} \quad (51)$$

During compression, the decreasing water level in the supply tank enhances the water flow rate through gravitational potential. However, as the test section water level rises, this gravitational effect diminishes, eventually becoming negative relative to the liquid level in the supply tank. This explains why the numerical model predicts a slower pressure increase compared with the experimental results. They can be also observed in water mass flowrate results in **Figure 12**.

4.2.2. Mass flowrate

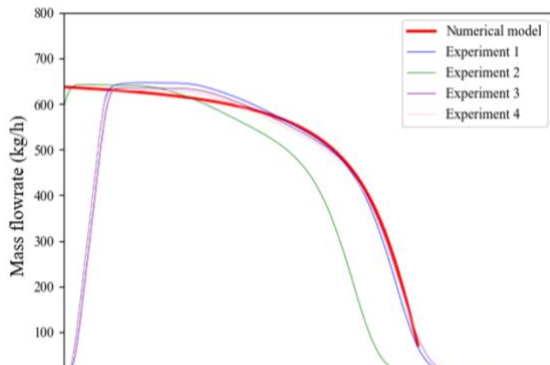


Figure 11: Pressure change during time. Experiment 1, 2, 3, and 4 are repeated experiments under same experiment conditions

There are experiment data process mistakes that the team are not setting flowrate starting time correctly for experiment 1, 3, and 4. For experiment 2, it still needs to be discussed. As mentioned before, the gravitational potential does bring positive effects at the beginning of compression, so that test section air pressure can be increased faster where mass flowrate is slower than the numerical result from 0 to 8 seconds. Later, the negative gravitational potential from water tank leads to deeper pressure slope by comparing with its earlier time.

4.2.3. Temperature

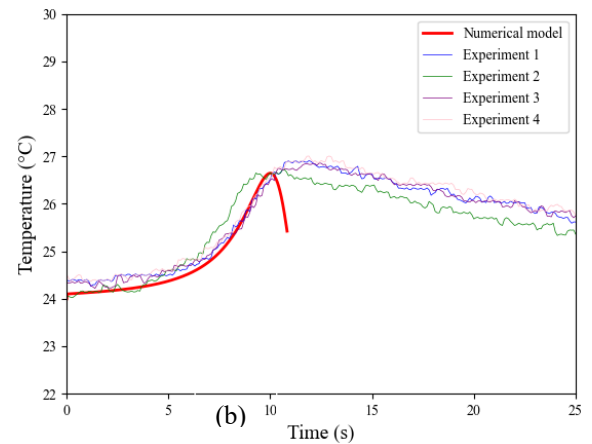
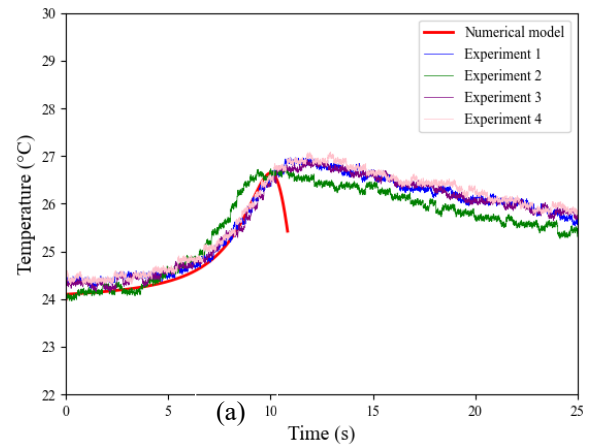


Figure 12: Pressure change during time. Experiment 1, 2, 3, and 4 are repeated experiments under same experiment conditions. (a) result is the original experimental data and (b) result is the smoothed result by eliminating the noise data.

In Figure 12, (a) result is the original data which is smoothed (b) by decrease data group size from the experiment group. However, this can be more accurate without decreasing group size by using Fourier transform in the future discussed. For the test section air temperature over time, experiment 2—as previously mentioned—starts with a higher initial pressure, and consequently a slightly lower initial temperature, approximately 0.3 °C lower.

This small deviation can be attributed to variations in environmental conditions.

From 0 to 10 seconds, the temperature profiles of all experiments follow the same trend as the pressure, showing a gradually increasing slope. During this phase, the temperature difference between the compressed air and the surrounding heat transfer media remains relatively small, allowing effective temperature control.

Each experiment reaches its peak temperature at a slightly different time. Experiment 2, with its lower initial temperature, reaches its peak of 26.5 °C first, at 9 seconds. This peak value is consistent with the numerical model prediction, though occurring slightly earlier because both share the same initial conditions. The earlier peak time indicates that experiment 2 reaches the pressure limit sooner, consistent with the observations in the pressure results.

For the other experiments, the peak temperature occurs at 10 seconds, with a slope that closely matches the numerical prediction. This suggests that identical initial temperatures lead to the same peak temperature, while identical initial pressures result in comparable temperature slopes, as supported by the comparison with the numerical results.

For the post-peak behavior, experiment 2 exhibits an earlier decreasing temperature slope, consistent with its earlier peak time. The other experiments, however, share a similar decreasing temperature slope. During this period, compression continues at nearly constant pressure while the air temperature gradually decreases. Without additional compression, the test section pressure tends to decline as the temperature decreases. Ultimately, this process yields compressed air characterized by relatively low temperature and high pressure.

In contrast, the numerical results show a faster rate of temperature decrease. This does not imply poor heat transfer efficiency in the experiments, but rather a limitation of the analytical model. Specifically, when applying the ideal gas law, the predicted pressure is greater than the actual measured pressure, and the predicted volume is smaller than the actual volume. This indicates that the model overestimates the air-wall convection effects, leading to a higher apparent heat transfer efficiency compared with the experimental results.

4.2.3. Heat flux

In the equation (57), the heat flux is defined by change in temperature and pressure. During compression, at the beginning, temperature negative change is getting larger and brings the first term to have negative result, and pressure positive change is getting larger and brings the second term to have negative result. Later, they both get decreased since the input pressure meets the limit, so the slope is getting flat. After the peak, temperature negative

change is decreasing, and pressure change is zero. They bring slope up and going back to no heat transfer trend.

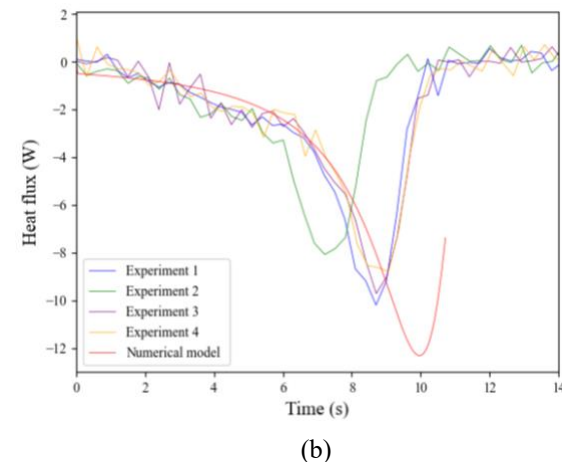
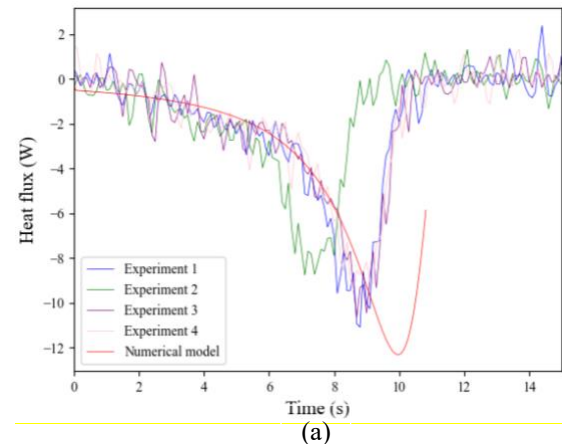


Figure 23: Heat flux during time. Experiment 1, 2, 3, and 4 are repeated experiments under same experiment conditions. (a) result is the original experimental data and (b) result is the smoothed result by eliminating the noise data.

In **Figure 13**, experiment 2 is finishing heat transfer from air to wall early with low peak. Experiment 2 with high initial pressure is meeting its limit pressure early, so heat transfer cannot last long as expected. Again, as mentioned before, the experiment results have unexpected and better compression speed at the beginning, so those results have early peak. The numerical model overestimates convection efficiency which leads to better performance by having lower peak.

5 . Discussion

From the perspectives of pressure, mass flow rate, temperature, and heat flux, the numerical model can capture the general slope trends and thermodynamic behaviors. However, due to the assumptions of ideal gas behavior and the simplified treatment of gravitational potential, the model predicts a slower

pressure slope compared with the experimental results and overestimates heat transfer efficiency. These two limitations should be addressed in future research.

Reflecting on the overall design, although only the compression process was simulated and experimentally executed, the use of a plate heat exchanger as a compressor to mitigate temperature fluctuations during compression is shown to be feasible based on the experimental results. A potential direction for future work is to replace the compressor as the source of input pressure with a water pump, which could enhance efficiency, improve operational flexibility, and eliminate constraints from input energy. In the long term, a complete compression–expansion nI-CAES system could be developed to evaluate round-trip efficiency and overall performance. Furthermore, comparative studies using different types of plate heat exchangers and experimental setups should be conducted. Finally, a two-dimensional analytical model could be developed and simulated to provide further insights into heat transfer mechanisms during both compression and expansion.

6 . Conclusion

This report goes through the introduction of various types of energy storage system including pumped hydro-electricity storage (PHES), hydrogen electricity storage (HES), electro chemical electricity storage (ECES), thermo-mechanical electricity storage (TMES), pumped thermal electricity storage (PTES), liquid air electricity storage (LAES), and compressed air electricity storage (CAES). By considering storage duration and scale, CAES can be considered as a good candidate for inter-seasonal and kWh scale needs. A technique and development introduction of existing CAES is made to discuss adiabatic, diabatic, and near-isothermal CAES. Due to the consideration of efficiency and sustainability, near-isothermal CAES has great potential to save energy loss and enhance efficiency by developing mechanism of compression. Liquid piston technology is the fundamental concept of nI-CAES, and its mechanism is explained. By considering the structure and advantage of heat transferring efficiency of plate heat exchanger, a design to transform plate heat exchanger to compressor is purposed in this report. A further discussion and design about making the system to be reversible compressor is sketched and discussed.

To examine numerically this design's performance, analytical model and numerical model are built to predict experiment result and system performance in 2 independent and 1-dimensional scales by using energy and mass conservation equations. A semi-infinite axial thermal boundary layer model, finite-domain solution for the gas-to-

plate lateral heat transfer model, and wall response model are built for analytical method. A numerical method including mass flowrate function and heat flux function is built and used to predict experiment result.

After simulating numerical model, mass flowrate of liquid, temperature, and pressure of test section air are plotted in **Figure 7, 8, and 9**. By comparing with experiment results from Clean Energy Lab's student, an other MSc, Yitong Han, the perspectives of pressure, mass flow rate, temperature, and heat flux, the numerical model successfully captures the overall slope trends and key thermodynamic behaviors. Nevertheless, the assumptions of ideal gas behavior and the simplified treatment of gravitational potential lead to a slower predicted pressure slope compared with the experimental results and an overestimation of heat transfer efficiency. Addressing these limitations will be an important direction for future research.

Reflecting on the overall design, although only the compression process was simulated and experimentally validated, the use of a plate heat exchanger as a compressor to mitigate temperature fluctuations during compression has been demonstrated to be feasible. Future work could explore replacing the compressor with a water pump as the input pressure source, thereby enhancing efficiency, improving operational flexibility, and removing constraints associated with input energy. In the longer term, a complete compression–expansion nI-CAES system could be developed to evaluate round-trip efficiency and overall performance. Additional research should also include comparative studies using different plate heat exchanger types and experimental configurations. Finally, the development and simulation of a two-dimensional analytical model would provide deeper insights into the heat transfer mechanisms during both compression and expansion.

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